

# 1.NEURO DRILL

## TECHNICAL SPECIFICATION – HIGH-SPEED NEUROSURGICAL DRILL SYSTEM

**Item:** Electric High-Speed Neurosurgical Drill System

**Intended Use:** Neurosurgery / Cranial Surgery / Spine Surgery / ENT

| Sl. No. | Technical Parameter            | Minimum Tender Requirement                                                                              |
|---------|--------------------------------|---------------------------------------------------------------------------------------------------------|
| 1       | Type                           | Electric high-speed surgical drill system suitable for neurosurgical procedures                         |
| 2       | Application                    | Suitable for drilling, cutting, shaping and removal of hard tissue, bone and bone replacement materials |
| 3       | Motor Speed                    | Variable speed range of <b>minimum 200 to 75000 RPM</b>                                                 |
| 4       | Maximum Mechanical Performance | <b>Power not less than 120w</b>                                                                         |
| 5       | Torque                         | <b>Minimum 2.2 Ncm</b>                                                                                  |
| 6       | Speed Control                  | Variable and precise speed control through foot control                                                 |
| 7       | Irrigation                     | Integrated irrigation system with flow rate of <b>at least 100 mL/min at 100%</b>                       |
| 8       | Drill Attachments              | Straight and angled drill attachments of different lengths                                              |
| 9       | Craniotome                     | Complete craniotome attachment suitable for craniotomy procedures                                       |
| 10      | Craniotome Guard               | Suitable protective/dural guard system                                                                  |
| 11      | Cutter/Burr System             | Universal cutter/burr system compatible with multiple surgical applications                             |
| 12      | Cutter Exposure                | Variable cutter exposure/adjustable working length                                                      |
| 13      | Attachment Coupling            | Quick <b>single locking system</b>                                                                      |
| 14      | Shaft Diameter                 | Slim shaft design suitable for microsurgical and minimally invasive approaches                          |
| 15      | Saw Attachments                | Suitable oscillating/sagittal/reciprocating saw attachments, as applicable                              |
| 16      | Perforator Compatibility       | Compatible with suitable cranial perforator system                                                      |
| 17      | Foot Control                   | Ergonomic foot-operated control for speed and irrigation control                                        |
| 18      | Power Supply                   | <b>220–240 V AC, 50–60 Hz</b>                                                                           |
| 19      | Permissible Voltage Variation  | ±10%                                                                                                    |

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|---------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 20      | Maximum Power Consumption | Approximately <b>170 VA or less</b>                                                                                                              |
| 21      | Control Unit              | Compact control unit with integrated motor control and irrigation management                                                                     |
| 22      | Dimensions                | Approximately <b>100 × 262 × 291 mm</b> or compact equivalent                                                                                    |
| 23      | Weight                    | Control unit including motor approximately <b>3.5 kg</b> or less                                                                                 |
| 24      | Electrical Safety         | Class II equipment with Type B applied part or equivalent safety classification                                                                  |
| 25      | Footswitch Protection     | Minimum <b>IPX8</b> or equivalent, where applicable                                                                                              |
| 26      | Sterilization             | Drill handpieces/attachments and reusable components shall be suitable for validated hospital sterilization procedures as per manufacturer's IFU |
| 27      | Noise & Vibration         | Low vibration and optimized thermal performance during continuous surgical use                                                                   |
| 28      | Accessories               | Complete set of required attachments, handpieces, foot control, irrigation accessories and compatible surgical instruments                       |
| 29      | Burs/Cutters              | Supply of a comprehensive range of compatible high-speed burs/cutters suitable for neurosurgical procedures                                      |
| 30      | Training                  | On-site installation, commissioning and comprehensive user training by authorized manufacturer's representative                                  |
| 31      | Warranty                  | Minimum <b>2 years comprehensive warranty</b> from date of installation/commissioning                                                            |
| 32      | Service Support           | Authorized service support available in India with trained service personnel                                                                     |
| 33      | Documentation             | User manual, service manual, technical specifications, electrical safety documents and preventive maintenance schedule to be supplied            |
| 34      | Regulatory Compliance     | Valid applicable regulatory approvals and certifications required for supply and use in India                                                    |
| 35      | Installation              | Installation, testing and commissioning shall be included in the quoted price                                                                    |

### Recommended Tender Accessories

The system should be supplied as a **complete working system**, including:

- Main electric high-speed drill control unit – 1 No.
- High-speed drill motor/handpiece – 1 No.
- Foot control/footswitch – 1 No.
- Straight drill attachments – minimum 3 Nos.
- Angled drill attachments – minimum 3 Nos.
- Craniotome attachment – 1 No.

- Craniotome protective/dural guards – suitable range
- Irrigation system with tubing/cannulae – 1 complete set
- Sterilization tray/container – 1 No.
- Complete set of reusable surgical attachments
- High-speed burs/cutters – assorted set
- Compatible cranial perforator – if included in tender requirement
- Required connecting cables and accessories
- User and service manuals

## **WARRANTY & AFTER SALE SUPPORT**

- **Comprehensive Warranty:** Minimum 3 years from the date of installation/commissioning.
- **Preventive Maintenance:** Minimum 4 preventive maintenance visits per year during the warranty period.
- **Spare Parts Support:** Availability and support for genuine spare parts for a minimum period of 10 years.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.